

MD KAMRUZZAMAN

Python Backend Developer | FastAPI | PostgreSQL | Docker | CI/CD

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Summary

Python backend developer with a B.Tech in Computer Science and Engineering (Machine Learning specialization). Built and deployed two backend and ML systems using FastAPI, PostgreSQL, Docker, CI/CD, MLflow, and Kubernetes. Experienced in REST API development, authentication, database design, automated testing, deployment, monitoring, and ML model serving. Focused on clean architecture, reliability, and test-driven backend development.

Technical Skills

Languages:	Python, SQL, Bash
Backend:	FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic
Databases:	PostgreSQL, MySQL
Auth & Security:	JWT, bcrypt, OAuth2
DevOps & Cloud:	Docker, Docker Compose, Kubernetes, GitHub Actions, Git, Linux, Render, Neon
Testing & Quality:	pytest, FastAPI TestClient, Ruff
ML / MLOps:	scikit-learn, MLflow, NumPy, Pandas
Observability:	Prometheus, Grafana, Evidently

Projects

Expense Tracker API

[Code](#) | [Live Demo](#)

- Built and deployed a REST API for personal expense tracking using FastAPI with JWT authentication, bcrypt password hashing, and user-scoped data isolation enforced at the query layer.
- Designed a PostgreSQL schema with SQLAlchemy ORM and managed database migrations with Alembic.
- Implemented pagination, category and amount filtering, Pydantic validation, structured error handling, and correct HTTP status codes.
- Achieved 91% test coverage with pytest across authentication, CRUD operations, authorization, and edge cases.
- Containerized with Docker and set up GitHub Actions for automated testing and Ruff linting.
- Deployed on Render with PostgreSQL on Neon, exposing a public Swagger/OpenAPI interface.

ML Expense Platform

[Code](#) | [Live Demo](#)

- Built an end-to-end ML serving platform that predicts expense categories from free-text descriptions.
- Developed a scikit-learn TF-IDF and Logistic Regression pipeline and tracked four MLflow experiments with regularization parameter sweeps.
- Built a FastAPI `/predict` endpoint with Pydantic validation and lazy model loading for inference.
- Containerized with Docker and wrote a Kubernetes deployment with two replicas, liveness and readiness probes, and self-healing behavior.
- Added Prometheus metrics for prediction counts by category and prediction latency, with Grafana dashboards for monitoring.
- Implemented Evidently-based data drift detection and validated it by injecting simulated production data (50% of columns flagged as drifted).

Education

Lovely Professional University, Punjab, India

2025

B.Tech in Computer Science and Engineering, Machine Learning Specialization

Certifications

- | | |
|--|------------------------|
| Meta Back-End Developer — Coursera | Verify |
| Natural Language Processing Specialization — DeepLearning.AI, Coursera | Verify |
| Python Data Structures — University of Michigan, Coursera | Verify |